

Business Analytics – SAS Base

Lecture 6: Join and Merge in SAS

Aim of the lecture

- Show you how to merge two datasets using merge function
- To do that, you also need to know how to use sort function

Agenda

- **What does merge do?**
- Functions needed to merge the datasets
- Recap

In real life, data is distributed across different tables with the **link variable**

Table 1

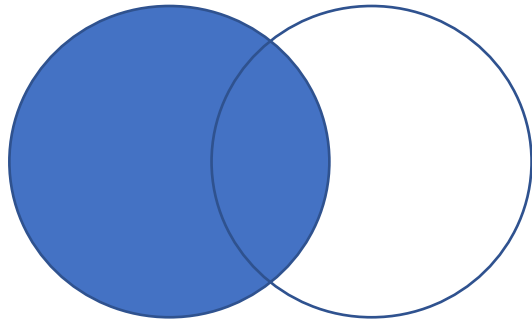
Order ID	Order time
1001	10/5/2017
1002	10/5/2017
1004	10/7/2017
1003	10/6/2017
1005	10/8/2017

Table 2

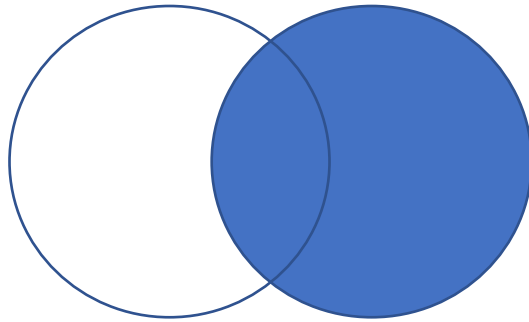
Order ID	Fulfillment time
1001	10/6/2017
1004	10/9/2017
1002	10/7/2017
2001	10/15/2017

There are four types of merge/join based on the needs of the analysis

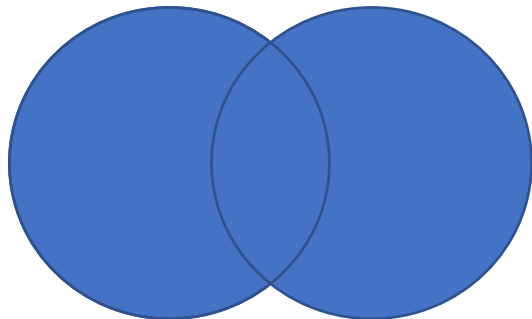
Left Join



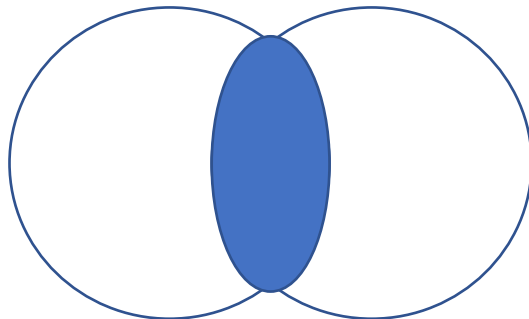
Right Join



Full Join



Inner Join



**Let us use
some
examples to
understand
the difference
between
different
types**

Which join would you use?

EXAMPLE 1

Dataset A has all the episodes played in Sep on Netflix with play time and episode ID;

Dataset B has the detailed info of the all of the episodes collected in Netflix with episode ID and title, genre, etc.

a) To find the detail of the episodes played in Sep, which join do you use?

b) To find the information regarding how long each episodes collected in Netflix has been played in Sep (and 0 if it was not played), which join do you use?

Which join would you use?

EXAMPLE 2

Dataset A has the credit history information from TransUnion;

Dataset B has the credit history information from Equifax

a) To find the credit score from either TransUnion or Equifax, which join do you use?

b) To find the information only for applicants with information from both TransUnion and Equifax, which join do you use?

Agenda

- What does merge do?
- **Functions needed to merge the datasets**
- Recap

We will go over the following functions

Function	What does it do?
SORT	Sort the data based on selected variable(s). Needed for merge to work
MERGE	Merge different datasets together based on the link
Multiple SET	Can also be used to combine datasets

SORT can be used to sort the dataset based on the selected variable(s)

NODUPKEY Option

```
PROC SORT DATA = dataset NODUPKEY;  
BY variable(s);  
RUN;
```

NODUP

```
PROC SORT DATA = dataset NODUP;  
BY variable(s);  
RUN;
```

MERGE can be used to implement different types of join

INNER JOIN

```
DATA output_data;  
MERGE data_a (IN = x) data_b(IN = y);  
BY variable(s);  
IF x AND y;  
RUN;
```

LEFT JOIN

```
IF x;
```

RIGHT JOIN

```
IF y;
```

FULL JOIN

```
no IF statement
```

Multiple set statements can be used to combine datasets as well

APPEND/CONCATENATE TWO DATA SETS

```
DATA output_data;  
SET data_a data _b  
RUN;
```

ADDING COLUMN FROM THE OTHER DATASET

```
DATA output_data;  
SET data_a;  
SET data_b;  
RUN;
```

Recap: Now you know how to combine different datasets in SAS

- Merge can be used to join datasets together (datasets need to be sorted)
- Multiple set statements can be useful as well to combine the datasets

*Thank
you*

